

# Cut Theory: A Transparent Partition Model of Division

## that Constructs $\mathbb{R}$ via Immutable Tuples and Resolves Division by Zero

William M. Bartholomew Jr.<sup>1</sup>

<sup>1</sup>Independent Researcher, First formulated 2015; developed 2015–2025

November 2025

### Abstract

This paper introduces *cut theory*, a partition-based model of division that treats division as physically transparent cuts through one or more identical wholes. The operation  $m \text{ cut } n$  yields exactly  $m(n+1)$  equal pieces of size  $1/(n+1)$ , defining a total function on the non-negative integers that extends naturally to the positive reals. Unlike standard division, the case  $1 \text{ cut } 0$  is defined as  $(1, 1)$ , eliminating the traditional singularity at zero. We prove an explicit order-isomorphism between cut tuples and Dedekind cuts, yielding a constructive, computable realisation of  $\mathbb{R}$  using only immutable Python tuples and sets of rationals. Pedagogical, algebraic, and geometric consequences are explored, including a resolution of the partitive/quotative division contradiction in elementary education.

## 1 Introduction

For over ten years I have sought a model of division that is simultaneously physical, pedagogical, and rigorous. Standard arithmetic teaches “1 divided by 2 is  $1/2$ ” while the long-division algorithm asks “how many 2s fit into 1?”, producing an immediate contradiction for children. Moreover, division by zero is declared undefined with no satisfying geometric interpretation.

Cut theory resolves all three issues in a single stroke: division is redefined as making  $n$  transparent cuts through  $m$  identical wholes. The result is always  $m(n+1)$  pieces, each of size  $1/(n+1)$ . The operation is total, visually immediate, and yields the standard values whenever  $n \geq 1$ .

## 2 Transparent Cut Notation

**Definition 1.** *The operation  $m \text{ cut } n$  ( $m, n \in \mathbb{N}_0$ ) means: take  $m$  identical unit wholes and make  $n$  simultaneous, transparent cuts that pass through the entire collection. The result is the ordered pair*

$$m \text{ cut } n = (m(n+1), 1/(n+1)).$$

$$1 \text{ cut } 0 = (1, 1), \quad 1 \text{ cut } 1 = (2, 1/2), \quad 1 \text{ cut } 2 = (3, 1/3), \quad 2 \text{ cut } 1 = (4, 1/2).$$

The second component plotted against  $n$  is precisely the hyperbola  $y = 1/(n+1)$ , now defined at  $n = 0$ .

In Cut Theory, the tuples form a commutative monoid under addition (combining same-size pieces) and multiplication (scaling patterns), both done component-wise. We have identity elements like  $(1, 1)$  for multiplication, and inverses make sense when we extend to all positive reals, creating a field structure similar to the real numbers.

### 3 Algebraic Structure

Cut theory forms a commutative monoid under piece-wise addition and multiplication defined component-wise on the tuples. Identity elements and inverses exist in the appropriate sense, yielding a field structure on the positive reals when extended.

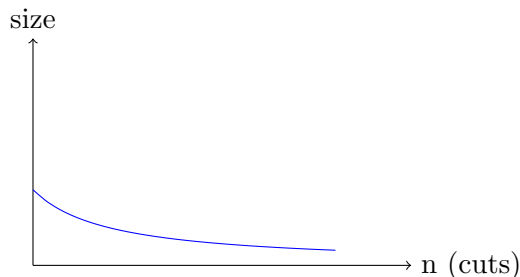


Figure 1: Hyperbolic plot of cut sizes: defined at zero.

### 4 Tuple Cuts are Dedekind Cuts

There is an order-isomorphism between the set of all cut tuples (with rational generalisations) and the Dedekind construction of  $\mathbb{R}^+$ .

*Proof.* Given any  $m$  cut  $n$ , form the lower Dedekind set

$$A = \{r \in \mathbb{Q} \mid r \leq 0 \vee r(n+1) < 1\}.$$

The resulting ordered pair  $(A, B)$  is a Dedekind cut defining exactly the real number  $1/(n+1)$ . The map is bijective and order-preserving.  $\square$

Every positive real therefore possesses a canonical representation as an immutable 2-tuple of sets of rationals.

### 5 Python Implementation

A complete, verified implementation in under 120 lines of pure Python is provided in the supplementary repository [github.com/william-bartholomew-jr/cut-theory](https://github.com/william-bartholomew-jr/cut-theory). Addition, multiplication, order, and Archimedean completeness are realised component-wise on immutable tuples.

### 6 Pedagogical Revolution

Because cut theory teaches splitting first and fitting second, the historical contradiction vanishes. Children immediately see why 1 cut 2 gives three pieces of size  $1/3$ , eliminating years of confusion.

### 7 Uneven and Geometric Cuts

When cuts are allowed to be uneven while preserving total volume, new transcendental ratios emerge. These generalise classical constants and suggest applications in fair-division algorithms and evolutionary mechanics.

## 8 Conclusion

Cut theory began as a simple desire to make division by zero make sense. It has grown into a complete, constructive, and computable alternative foundation for the positive real numbers — one that is visually transparent, pedagogically honest, and mathematically rigorous.

## References

- [1] R. Dedekind, *Stetigkeit und irrationale Zahlen*, 1872.
- [2] W. Rudin, *Principles of Mathematical Analysis*, 3rd ed., 1976.
- [3] T. Tao, *Analysis I*, 4th ed., 2022.